Logistic Regression Exercise

Scenario

You're helping a Cameroonian bank predict whether a loan applicant will **default** (fail to repay) based on two features: their monthly income and existing debt amount.

```
import numpy as np
import pandas as pd

data = {
    'monthly_income': [80000, 150000, 45000, 200000, 60000,
                        250000, 90000, 300000, 55000, 180000,
                        70000, 220000, 40000, 160000, 95000],
    'existing_debt': [50000, 20000, 60000, 10000, 55000,
                       15000, 45000, 5000, 65000, 25000,
                       48000, 12000, 70000, 18000, 40000],
    'defaulted': [1, 0, 1, 0, 1,
                  0, 1, 0, 1, 0,
                  1, 0, 1, 0, 0]  # 1 = defaulted, 0 = repaid
}

df = pd.DataFrame(data)
print(df)
```

Tasks

1. Explore the data

- Print df.describe() to see the summary statistics
- Count how many people defaulted vs repaid using df['defaulted'].value_counts()

2. Prepare the data

- Split into features (X) and target (y)
- Use train_test_split (80% train, 20% test)

3. Train a Logistic Regression model

```
from sklearn.linear_model import LogisticRegression
```

- Fit the model on the training data
- Print the model's coefficients (model.coef_) and intercept (model.intercept_)

4. Make predictions

- Predict on the test set using model.predict(X_test)
- Also try model.predict_proba(X_test) — this shows the actual probability for each class instead of just 0 or 1

5. Interpret one applicant

- Pick a new hypothetical applicant: income = 100,000, debt = 40,000
- Predict whether they'd default, and print the probability